Report Exercize 2

Image Preprocessing

The image was first converted to grayscale using *im2gray* to simplify the enhancement process, since contrast operations are more meaningful in a single intensity channel. It was then normalized to double precision in the range [0, 1] using *im2double*, ensuring consistency for all subsequent operations.

ROI Selection

The user was asked to manually select 12 polygonal regions of interest (ROIs) (the number of ROIs was set in advance to make sure the user selected all of the squares inside the circle), corresponding to the internal squares.

This interactive selection was implemented using *drawpolygon*, which allows precise localization and control of each area to be enhanced.

Each ROI was stored as a logical mask (*createMask*) to enable selective processing of pixels.

Contrast Enhancement

For each ROI, the contrast was enhanced using the contrast stretching method via the function *imadjust(x, stretchlim(x, [0.01 0.99]), [])*

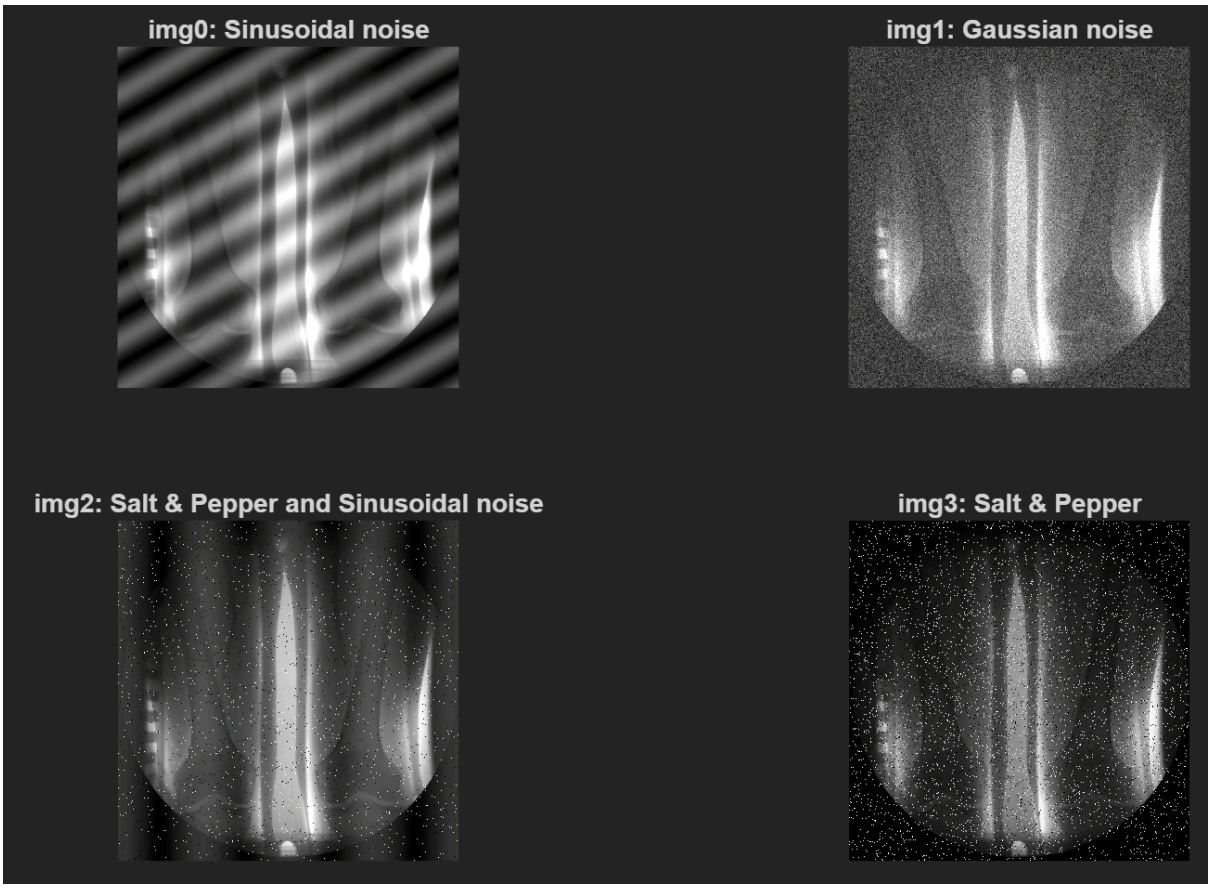
This operation linearly scales the pixel intensities between the 1st and 99th percentile, effectively enhancing local contrast while avoiding saturation due to extreme outliers.

The enhanced pixel values were then reassigned only to the corresponding ROI, ensuring that the rest of the image remained unaltered.

Visualization

The final enhanced image was displayed alongside the original one for direct visual comparison. This step helps evaluate the effectiveness of the enhancement, showing improved local contrast within the selected squares.

Report Exercize 4



In the figure, the 4 images contained in h4_2025.mat are shown and, as indicated by the title, the noises affecting each image are as follows:

1. img0 has sinusoidal noise with spatial frequencies along x and y equal to 4/N cycles per sample and 6/N cycles per sample, respectively.

2. img1 has Gaussian noise.

3. img2 has two types of noise:
    a. a sinusoidal noise along the x-axis equal to 4/N cycles per sample,
    b. low-density salt-and-pepper noise.

4. img3 has high-density salt-and-pepper noise.

Relevant parameters of each noise

A Gaussian noise pattern is one of the most commonly encountered forms of noise in digital images and other signals. It arises when numerous independent, minute random fluctuations, from sources such as sensor imperfections, thermal effects, or transmission errors, are summed together and alter the pixel values in an image. This noise is conventionally modeled using a normal (Gaussian) probability distribution, so that most changes are small and centered near a mean value, while large deviations are rare and symmetrically distributed around that mean. The two primary parameters

describing Gaussian noise are the mean (μ), which represents the average bias or shift introduced to the pixel intensities, and the standard deviation (σ), which quantifies the spread or intensity of these random variations around the mean. As a result, Gaussian noise creates a smooth, grainy randomness across the entire image, with variations in pixel intensity that are statistically predictable and evenly dispersed rather than concentrated or isolated.

Salt-and-pepper noise manifests as abrupt and isolated spikes in intensity, randomly setting some pixels to the lowest (0) or highest (255) values in the image's display range. This effect is visually reminiscent of grains of salt (bright spots) and grains of pepper (dark spots) scattered across the image. The fundamental parameter for this noise is its density (d), which defines the proportion of pixels that are affected and replaced by these extreme values. Unlike Gaussian noise, which impacts all pixels in a subtle and continuous manner, salt-and-pepper noise creates conspicuous, isolated points of maximum or minimum intensity. Each noisy pixel is selected at random, making the effect entirely independent of spatial location or the properties of neighboring pixels.